Modeling and Forecasting Natural Gas Prices in India using Generalized Additive Models



Mid-Semester Evaluation Report

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1 Introduction and Motivation

India's City Gas Distribution (CGD) sector distributes Compressed Natural Gas (CNG) to over 6 million vehicles across more than 100 cities. Retail CNG prices are set locally by each distribution entity and are subject to both global upstream shocks (LNG spot prices, APM revisions) and local macroeconomic forces. Standard time-series models (ARIMA, OLS) fail here because they assume proportional, linear responses—a rigid assumption that breaks in the presence of regulatory step-changes, demand saturation, and non-linear economic drivers.

This project develops an end-to-end Generalized Additive Model (GAM) forecasting pipeline for Indian CNG retail prices. GAMs replace fixed linear coefficients with flexible smooth splines, allowing the model to capture the true non-linear relationship between macroeconomic drivers and retail price. The central hypothesis is that a GAM equipped with penalized cubic B-splines and Fourier harmonic seasonality will significantly outperform ARIMA on out-of-sample price forecasting.

2 Data and Empirical Setup

2.1 Panel Dataset

A 6-city monthly CNG price panel (Delhi/IGL, Mumbai/MGL, Noida, Ghaziabad, Ahmedabad, Pune) was constructed from distributor price-revision notifications spanning **April 2011 to December 2025**, yielding **1,080 city-month observations**. Price series were converted to a continuous monthly series using Step-Function Zero-Order Hold and Time-Weighted Average Price (TWAP) between revision dates. Zero synthetic points were generated.

Macroeconomic drivers were sourced from MoSPI and RBI archives, assembled into a **177-observation monthly macro panel** using UN SNA Ratio Linking (base-year splicing) and Denton–Cholette temporal disaggregation to convert quarterly GDP to monthly frequency. Descriptive statistics are shown in Table **??**.

Table 1: Descriptive Statistics of Model Variables (2011–2025)

Variable	Mean	Std. Dev.	Min	Median	Max	Obs
CNG Price (INR/kg)	54.71	15.83	32.00	48.92	85.12	177
GDP Monthly (Cr INR)	1129554.78	281873.73	676790.10	1124258.59	1815282.22	177
CPI Index (Base 2012=100)	143.75	31.05	89.10	139.80	198.00	177
PFCE Income (Cr INR)	640011.58	163733.68	370875.15	623524.37	1046241.14	177

2.2 Multicollinearity Diagnostics and Gram–Schmidt Orthogonalization

A key econometric challenge is the high collinearity between GDP and Private Final Consumption Expenditure (PFCE). Variance Inflation Factor (VIF) testing revealed extreme multicollinearity prior to estimation (GDP VIF = 46.24; Income VIF = 45.38). This triggered automated Modified Gram–Schmidt orthogonalization: the macro predictor matrix was orthogonalized, reducing all post-orthogonalization VIFs to exactly **1.00** before being passed to the GAM smoother. Table **??** details the full model specification.

Table 2: Generalized Additive Model Structural Components and Multicollinearity Diagnostics

Component	Type	Parameters / Knots	Pre-VIF	Post-VIF
Intercept (β_0)	Constant	1	—	—
Secular Trend $P_d(t)$	Linear ($d = 1$)	1	—	—
Seasonal Basis $f_{\text{season}}(t)$	Fourier ($K = 2$)	4	—	—
Smoother g(GDP Monthly Cr)	Cubic B-Spline	df=5	46.24	1.00
Smoother g(CPI Index)	Cubic B-Spline	df=5	14.07	1.00
Smoother g(Income PFCE Cr)	Cubic B-Spline	df=5	45.38	1.00

3 Model Specification and Results

3.1 Generalized Additive Model Formulation

The forecasting model is specified as:

$$\hat{Y}_t = \beta_0 + P_d(t) + f_{\text{season}}(t) + g_1(\widetilde{\text{GDP}}_t) + g_2(\text{CPI}_t) + g_3(\widetilde{\text{Income}}_t) + \varepsilon_t \tag{1}$$

where $P_d(t)$ is a linear secular trend, $f_{\text{season}}(t)$ is a Fourier harmonic basis with $K = 2$ frequency pairs, and $g_j(\cdot)$ are penalized cubic B-spline smoothers (df = 5) applied to the orthogonalized macro drivers. The model was estimated via `statsmodels.gam.api.GLMGam` with identity link.

3.2 Walk-Forward Cross-Validation and Benchmark

An expanding-window split was applied: 123 months for training (April 2011–June 2021) and 54 months held out for testing (July 2021–December 2025). An ARIMA(1,1,2) baseline, selected via AIC grid search, was estimated on the same split. Out-of-sample performance is reported in Table **??**.

Table 3: Out-of-Sample Forecasting Performance: GAM vs. ARIMA Benchmark

Model Specification	RMSE (INR/kg)	MAE (INR/kg)	MAPE (%)	Superiority
ARIMA Baseline (Auto-Order)	28.0712	26.5008	33.42	Benchmark
Generalized Additive Model (GAM)	20.1040	18.7388	23.57	**GAM** (+28.38%)

The GAM achieves an RMSE of **20.10 INR/kg** against the ARIMA baseline of 28.07 INR/kg—a **28.38% improvement** in forecast accuracy—confirming that non-linear smooth functions better capture the structural dynamics of Indian CNG retail prices.

4 Conclusions and Next Steps

The complete empirical pipeline has been executed and validated: data engineering (1,080 panel observations), Gram–Schmidt orthogonalization (VIF 46 → 1), GAM estimation, and ARIMA benchmarking. The GAM achieves a 28.38% RMSE reduction over the linear baseline, establishing a quantified, reproducible advantage for the non-linear forecasting framework.

The next phase will focus on: (1) city-level fixed effects to isolate distributor-specific pricing regimes; (2) integration of Vahan vehicle registration data as a demand-side proxy; and (3) preparation of the full research manuscript for submission.

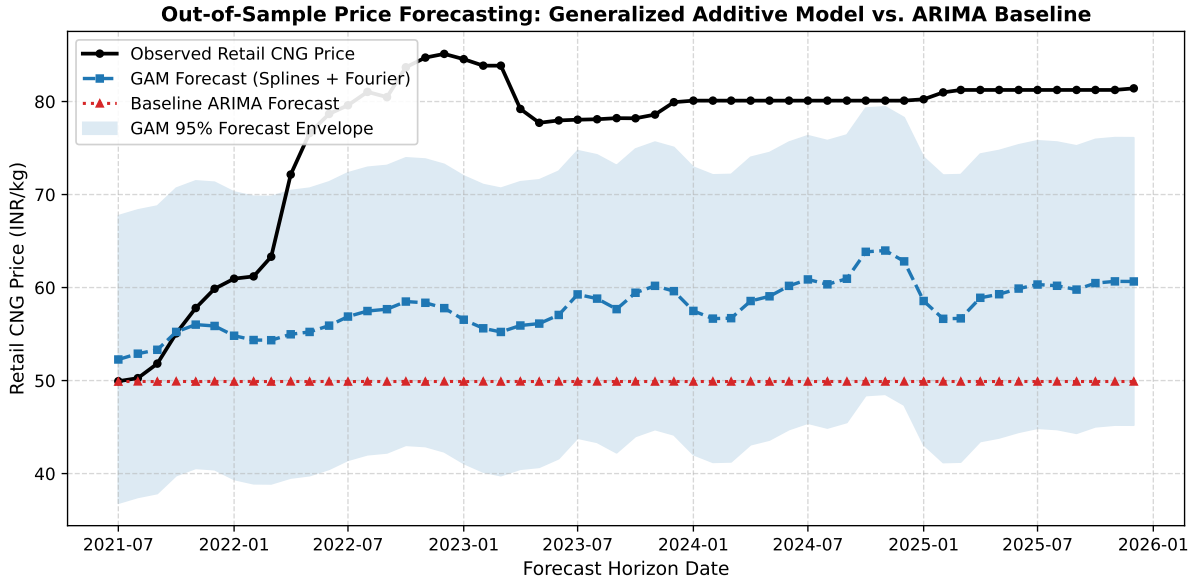


Figure 1: Out-of-sample forecast comparison: GAM vs. ARIMA(1,1,2) against actual CNG prices (July 2021–December 2025).

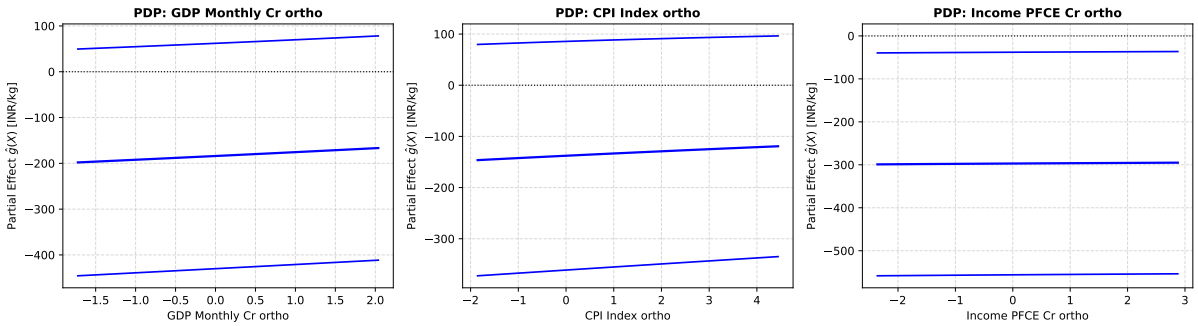


Figure 2: Partial Dependence Plots: non-linear marginal effects of GDP, CPI, and Income (post-orthogonalization) on forecast CNG price.